DISCO-TAB revision — table review

This file shows the three revised tables in the JBHI/IEEE two-column geometry. Table I (datasets), Table II (aggregate benchmark), Table III (DISCO-TAB vs. best competitor).

TABLE I
BENCHMARK DATASETS (23 TOTAL: 9 FROM THE ORIGINAL STUDY PLUS 10 CLINICAL TIME-TO-EVENT COHORTS FROM SURVSET AND 4 FROM TABARENA). TIME-TO-EVENT COHORTS USE THE EVENT INDICATOR AS A BINARY TARGET.

Dataset	**Domain**	**#Inst.**	**#Feat.**	**#Cls.**	**Source**
Heart Failure	Clinical	299	12	2	[**?**]
Breast Cancer	Clinical	569	30	2	[**?**]
Liver Disorders	Clinical	345	5	2	[**?**]
Parkinsons	Clinical	195	22	2	[**?**]
Obesity	Clinical	2111	16	7	[**?**]
Diabetes	Clinical	100000	8	2	[**?**]
German Credit	Finance	1000	20	2	[**?**]
Bank Marketing	Finance	45211	16	2	[**?**]
Census Income	Finance	48842	14	2	[**?**]
Melanoma	Clinical	205	5	2	[**?**]
PBC	Clinical	312	6	2	[**?**]
PBC3	Clinical	349	12	2	[**?**]
Retinopathy	Clinical	368	7	2	[**?**]
WHAS500	Clinical	461	16	2	[**?**]
GBSG2	Clinical	686	8	2	[**?**]
Colon	Clinical	885	9	2	[**?**]
GRACE	Clinical	998	5	2	[**?**]
TRACE	Clinical	1878	6	2	[**?**]
Framingham	Clinical	4699	7	2	[**?**]
Blood Transfusion	Clinical	533	4	2	[**?**]
QSAR Biodeg	Chemistry	1052	41	2	[**?**]
Maternal Risk	Clinical	452	6	3	[**?**]
Website Phishing	Security	724	9	3	[**?**]

TABLE II

BENCHMARK SUMMARY: MEAN OVER ALL DATASETS (WITH 95% CI) FOR EACH METRIC AND GENERATOR. **BOLD** = BEST PER ROW AMONG THE ESTABLISHED GENERATORS. ARROWS GIVE THE PREFERRED DIRECTION (MIA IS BEST NEAR 0.5). ESTABLISHED GENERATORS COVER ALL 23 DATASETS; LLM-ERA METHODS COVER THE SUBSET SHOWN IN THE #DATASETS ROW ('–' = NOT EVALUATED).

		Established generators										**LLM-era**			
	Metric	**DISCO-TAB**	**CTGAN**	**TVAE**	**GaussCop**	**TabDDPM**	**ADS-GAN**	**PATE-GAN**	**RTVAE**	**NFlow**	**Marg.**	**GReaT**	**RTF**	**TAEGAN**	**TAGAL**
	#datasets	23	23	23	23	23	23	23	23	23	23	4	1	6	3
Fidelity Utility (TSTR)	F1-macro ↑	**0.641 ±0.062**	0.470 ±0.061	0.403 ±0.128	0.543 ±0.057	0.603 ±0.074	0.620 ±0.047	0.423 ±0.041	0.578 ±0.050	0.538 ±0.057	0.409 ±0.047	0.454 ±0.084	0.795	0.683 ±0.123	0.658 ±0.278
	F1-weight. ↑	0.681 ±0.067	0.549 ±0.077	0.453 ±0.139	0.640 ±0.064	0.683 ±0.067	**0.687 ±0.054**	0.530 ±0.071	0.665 ±0.054	0.632 ±0.065	0.456 ±0.058	0.660 ±0.301	0.827	0.752 ±0.140	0.740 ±0.195
	Accuracy ↑	0.688 ±0.064	0.578 ±0.079	0.485 ±0.142	0.670 ±0.066	**0.714 ±0.062**	0.706 ±0.052	0.583 ±0.075	0.688 ±0.056	0.666 ±0.062	0.464 ±0.056	0.718 ±0.304	0.833	0.771 ±0.131	0.718 ±0.215
	JS div. ↓	0.059 ±0.020	0.088 ±0.032	0.132 ±0.026	0.047 ±0.028	0.084 ±0.057	0.070 ±0.022	0.174 ±0.051	0.232 ±0.068	**0.035 ±0.012**	0.208 ±0.045	0.049 ±0.014	0.013	0.073 ±0.050	0.055 ±0.095
	KS stat. ↓	0.169 ±0.038	0.217 ±0.044	0.267 ±0.036	**0.108 ±0.035**	0.120 ±0.056	0.183 ±0.031	0.349 ±0.056	0.248 ±0.042	0.112 ±0.016	0.389 ±0.062	0.140 ±0.078	0.052	0.152 ±0.073	0.146 ±0.198
	Helling. ↓	0.170 ±0.026	0.197 ±0.046	0.300 ±0.036	**0.125 ±0.038**	0.172 ±0.075	0.184 ±0.037	0.298 ±0.055	0.347 ±0.082	0.136 ±0.023	0.356 ±0.054	0.154 ±0.010	0.087	0.193 ±0.091	0.166 ±0.166
	Corr. ↑	0.908 ±0.029	0.821 ±0.055	0.844 ±0.034	**0.929 ±0.015**	0.886 ±0.033	0.913 ±0.018	0.797 ±0.054	0.865 ±0.041	0.862 ±0.039	0.818 ±0.055	0.894 ±0.050	0.939	0.903 ±0.036	0.868 ±0.127
Privacy	DCR ↑	0.338 ±0.077	0.556 ±0.134	0.255 ±0.072	0.399 ±0.094	0.763 ±0.521	0.357 ±0.086	0.612 ±0.164	0.334 ±0.060	0.404 ±0.101	**1.420 ±0.404**	0.364 ±0.445	0.684	0.401 ±0.233	0.470 ±0.779
	NNDR ↑	0.885 ±0.038	0.872 ±0.044	0.949 ±0.017	0.846 ±0.053	0.814 ±0.073	0.871 ±0.043	0.882 ±0.036	**0.953 ±0.022**	0.857 ±0.048	0.868 ±0.049	0.890 ±0.041	0.880	–	–
	MIA →.5	0.546 ±0.020	0.503 ±0.009	0.502 ±0.008	0.501 ±0.007	0.537 ±0.025	0.512 ±0.007	**0.499 ±0.004**	0.512 ±0.010	0.503 ±0.008	0.501 ±0.006	0.488 ±0.072	0.444	–	–
	Dup. ↓	0.045 ±0.049	**0.003 ±0.003**	0.029 ±0.041	0.006 ±0.009	0.062 ±0.090	0.020 ±0.032	0.008 ±0.014	0.019 ±0.030	0.016 ±0.027	0.003 ±0.005	0.000 ±0.000	0.000	–	–
Diversity	Precis. ↑	0.883 ±0.086	0.615 ±0.124	**0.896 ±0.091**	0.824 ±0.094	0.808 ±0.116	0.818 ±0.101	0.511 ±0.120	0.820 ±0.087	0.728 ±0.133	0.441 ±0.122	0.936 ±0.079	0.987	–	–
	Recall ↑	0.756 ±0.104	0.895 ±0.040	0.323 ±0.088	0.929 ±0.032	**0.967 ±0.021**	0.857 ±0.033	0.839 ±0.080	0.611 ±0.106	0.941 ±0.023	0.918 ±0.076	0.843 ±0.102	0.954	–	–
	Density ↑	0.974 ±0.175	0.414 ±0.109	**1.134 ±0.218**	0.682 ±0.118	0.930 ±0.256	0.806 ±0.198	0.341 ±0.146	0.950 ±0.221	0.681 ±0.198	0.227 ±0.101	1.031 ±0.178	1.152	–	–
	Coverage ↑	0.764 ±0.107	0.525 ±0.115	0.395 ±0.083	0.774 ±0.110	**0.876 ±0.112**	0.710 ±0.096	0.365 ±0.095	0.625 ±0.108	0.703 ±0.134	0.223 ±0.108	0.783 ±0.234	0.987	–	–
Detection	C2ST ↓	0.889 ±0.058	0.934 ±0.035	0.960 ±0.021	0.815 ±0.070	**0.726 ±0.089**	0.909 ±0.047	0.966 ±0.021	0.978 ±0.032	0.873 ±0.061	0.972 ±0.019	0.841 ±0.044	0.442	–	–
	MMD ↓	0.039 ±0.013	0.100 ±0.033	0.176 ±0.042	0.021 ±0.018	0.074 ±0.068	0.040 ±0.012	0.238 ±0.076	0.048 ±0.022	**0.019 ±0.005**	0.393 ±0.094	0.034 ±0.031	0.000	–	–
	Wass. ↓	0.233 ±0.038	0.414 ±0.087	0.444 ±0.059	**0.173 ±0.069**	0.583 ±0.410	0.263 ±0.038	0.664 ±0.118	0.314 ±0.055	0.178 ±0.014	1.755 ±0.678	0.216 ±0.046	0.094	–	–
SDMetrics	Quality ↑	0.629 ±0.080	0.561 ±0.063	0.548 ±0.074	**0.666 ±0.077**	0.662 ±0.101	0.631 ±0.082	0.495 ±0.063	0.554 ±0.073	0.624 ±0.072	0.462 ±0.056	0.499 ±0.216	0.438	–	–
	Shapes ↑	0.770 ±0.031	0.729 ±0.044	0.674 ±0.038	**0.828 ±0.023**	0.784 ±0.049	0.764 ±0.038	0.626 ±0.054	0.715 ±0.048	0.810 ±0.026	0.584 ±0.049	0.782 ±0.064	0.875	–	–
	Pairs ↑	0.488 ±0.154	0.394 ±0.117	0.423 ±0.136	0.504 ±0.153	**0.539 ±0.187**	0.498 ±0.155	0.364 ±0.111	0.393 ±0.134	0.438 ±0.133	0.339 ±0.107	0.216 ±0.447	0.000	–	–

TABLE III

DISCO-TAB VS. THE BEST ESTABLISHED COMPETING GENERATOR PER METRIC (MEAN OVER ALL DATASETS, WITH 95% CI). **BOLD** = WINNER.

	Metric	DISCO-TAB	Best comp.	Method
Fidelity/Utility (TSTR)	F1-macro ↑	**0.641** ±0.062	0.620 ±0.047	ADS-GAN
	F1-weight. ↑	0.681 ±0.067	**0.687** ±0.054	**ADS-GAN**
	Accuracy ↑	0.688 ±0.064	**0.714** ±0.062	**TabDDPM**
	JS div. ↓	0.059 ±0.020	**0.035** ±0.012	**NFlow**
	KS stat. ↓	0.169 ±0.038	**0.108** ±0.035	**GaussCop**
	Helling. ↓	0.170 ±0.026	**0.125** ±0.038	**GaussCop**
	Corr. ↑	0.908 ±0.029	**0.929** ±0.015	**GaussCop**
Privacy	DCR ↑	0.338 ±0.077	**1.420** ±0.404	**Marg.**
	NNDR ↑	0.885 ±0.038	**0.953** ±0.022	**RTVAE**
	MIA →.5	0.546 ±0.020	**0.499** ±0.004	**PATE-GAN**
	Dup. ↓	0.045 ±0.049	**0.003** ±0.003	**CTGAN**
Diversity	Precis. ↑	0.883 ±0.086	**0.896** ±0.091	**TVAE**
	Recall ↑	0.756 ±0.104	**0.967** ±0.021	**TabDDPM**
	Density ↑	0.974 ±0.175	**1.134** ±0.218	**TVAE**
	Coverage ↑	0.764 ±0.107	**0.876** ±0.112	**TabDDPM**
Detection	C2ST ↓	0.889 ±0.058	**0.726** ±0.089	**TabDDPM**
	MMD ↓	0.039 ±0.013	**0.019** ±0.005	**NFlow**
	Wass. ↓	0.233 ±0.038	**0.173** ±0.069	**GaussCop**
SDMetrics	Quality ↑	0.629 ±0.080	**0.666** ±0.077	**GaussCop**
	Shapes ↑	0.770 ±0.031	**0.828** ±0.023	**GaussCop**
	Pairs ↑	0.488 ±0.154	**0.539** ±0.187	**TabDDPM**